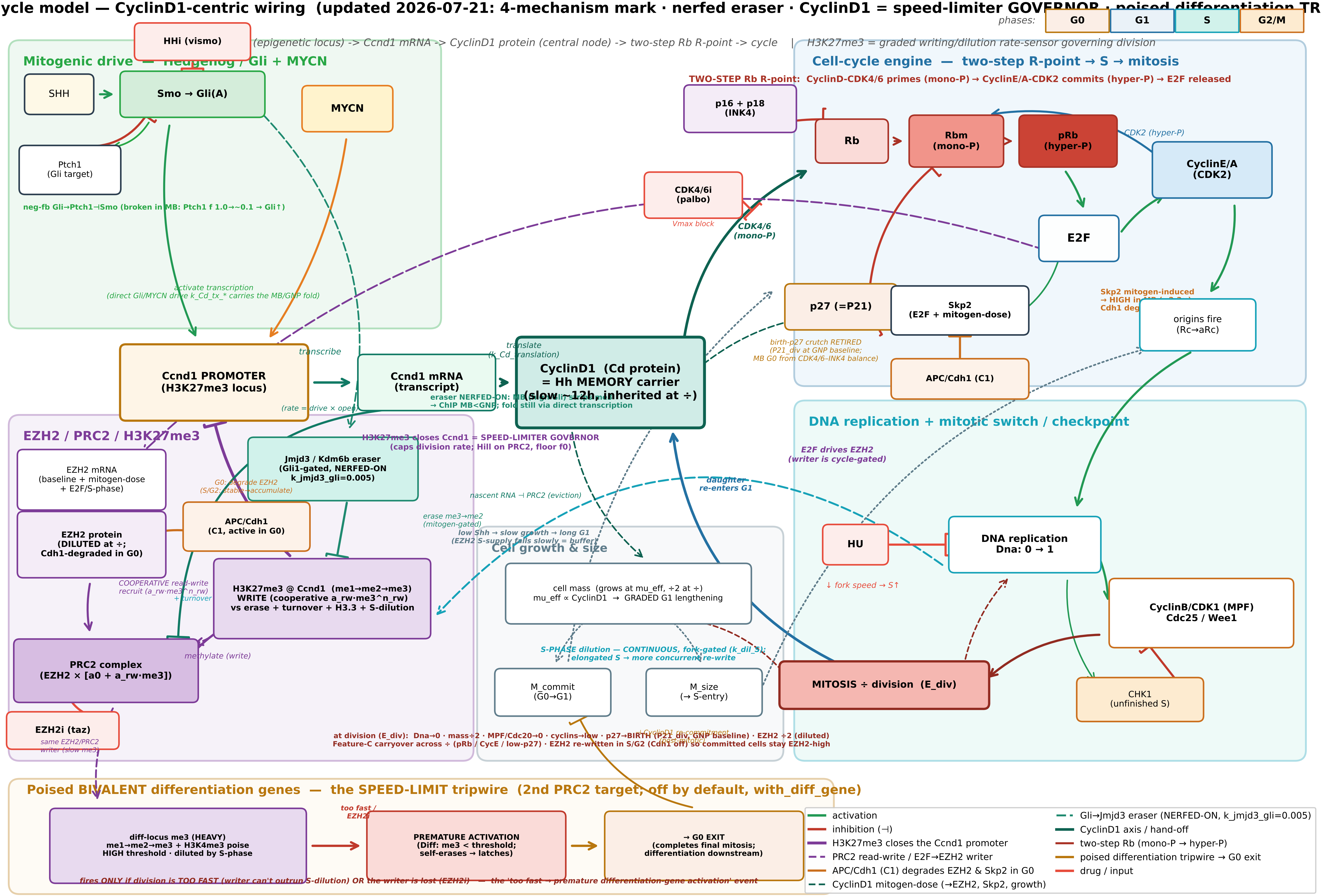
v44 cell-cycle model — CyclinD1-centric wiring (updated 2026-07-21: 4-mechanism mark · nerfed eraser · CyclinD1 = speed-limiter GOVERNOR · poised differentiation TRIPWIRE)



phases:
G0
G1
S
G2/M
(epigenetic locus) -> Ccnd1 mRNA -> CyclinD1 protein (central node) -> two-step Rb R-point -> cycle | H3K27me3 = graded writing/dilution rate-sensor governing division
HHi (vismo)
Mitogenic drive — Hedgehog / Gli + MYCN
SHH
Smo → Gli(A)
MYCN
Ptch1 (Gli target)
neg-fb Gli→Ptch1⊣Smo (broken in MB: Ptch1 f 1.0→~0.1 → Gli↑)
activate transcription (direct Gli/MYCN drive k_Cd_tx_* carries the MB/GNP fold)
Cell-cycle engine — two-step R-point → S → mitosis
TWO-STEP Rb R-point: CyclinD-CDK4/6 primes (mono-P) → CyclinE/A-CDK2 commits (hyper-P) → E2F released
p16 + p18 (INK4)
Rb
Rbm (mono-P)
pRb (hyper-P)
CDK2 (hyper-P)
CyclinE/A (CDK2)
CDK4/6i (palbo)
Vmax block
CDK4/6 (mono-P)
E2F
p27 (=P21)
Skp2 (E2F + mitogen-dose)
Skp2 mitogen-induced
→ HIGH in
Cdh1 deg
origins fire (Rc→aRc)
birth-p27 crutch RETIRED (P21_div at GNP baseline; MB G0 from CDK4/6–INK4 balance)
APC/Cdh1 (C1)
transcribe
Ccnd1 PROMOTER (H3K27me3 locus)
(rate = drive × open)
Ccnd1 mRNA (transcript)
translate (k_Cd_translation)
CyclinD1 (Cd protein) = Hh MEMORY carrier (slow ~12h, inherited at ÷)
eraser NERFED-ON: MB (hi Gli) strips me3 → ChIP MB<GNP; fold still via direct transcription
EZH2 / PRC2 / H3K27me3
H3K27me3 closes Ccnd1 = SPEED-LIMITER GOVERNOR (caps division rate; Hill on PRC2, floor f0)
EZH2 mRNA (baseline + mitogen-dose + E2F/S-phase)
Jmjd3 / Kdm6b eraser (Gli1-gated, NERFED-ON k_jmjd3_gli=0.005)
G0: degrade EZH2 (S/G2: stable→accumulate)
EZH2 protein (DILUTED at ÷; Cdh1-degraded in G0)
APC/Cdh1 (C1, active in G0)
nascent RNA ⊣ PRC2 (eviction)
erase me3→me2 (mitogen-gated)
low Shh → slow growth → long G1 (EZH2 S-supply falls slowly = buffer)
COOPERATIVE read-write recruit (a_rw·me3^n_rw) + turnover
H3K27me3 @ Ccnd1 (me1→me2→me3) WRITE (cooperative a_rw·me3^n_rw) vs erase + turnover + H3.3 + S-dilution
PRC2 complex (EZH2 × [a0 + a_rw·me3])
methylate (write)
EZH2i (taz)
same EZH2/PRC2 writer (slow me3)
daughter re-enters G1
E2F drives EZH2 (writer is cycle-gated)
DNA replication + mitotic switch / checkpoint
Cell growth & size
cell mass (grows at mu_eff, ÷2 at ÷) mu_eff ∝ CyclinD1 → GRADED G1 lengthening
S-PHASE dilution — CONTINUOUS, fork-gated (k_dil_S); elongated S → more concurrent re-write
M_commit (G0→G1)
M_size (→ S-entry)
HU
↓ fork speed → S↑
DNA replication Dna: 0 → 1
CyclinB/CDK1 (MPF) Cdc25 / Wee1
MITOSIS ÷ division (E_div)
CHK1 (unfinished S)
at division (E_div): Dna→0 · mass÷2 · MPF/Cdc20→0 · cyclins→low · p27→BIRTH (P21_div, GNP baseline) · EZH2 ÷2 (diluted)
Feature-C carryover across ÷ (pRb / CycE / low-p27) · EZH2 re-written in S/G2 (Cdh1 off) so committed cells stay EZH2-high
⊣ CyclinD1 re-commitment (post-mitotic)
Poised BIVALENT differentiation genes — the SPEED-LIMIT tripwire (2nd PRC2 target; off by default, with_diff_gene)
diff-locus me3 (HEAVY) me1→me2→me3 + H3K4me3 poise HIGH threshold · diluted by S-phase
too fast / EZH2i
PREMATURE ACTIVATION (Diff: me3 < threshold; self-erases → latches)
→ G0 EXIT (completes final mitosis; differentiation downstream)
fires ONLY if division is TOO FAST (writer can't outrun S-dilution) OR the writer is lost (EZH2i) — the 'too fast → premature differentiation-gene activation' event
activation
inhibition (⊣)
H3K27me3 closes the Ccnd1 promoter
PRC2 read-write / E2F→EZH2 writer
APC/Cdh1 (C1) degrades EZH2 & Skp2 in G0
CyclinD1 mitogen-dose (→EZH2, Skp2, growth)
Gli→Jmjd3 eraser (NERFED-ON, k_jmjd3_gli=0.005)
CyclinD1 axis / hand-off
two-step Rb (mono-P → hyper-P)
poised differentiation tripwire → G0 exit
drug / input